**1** **(a)** **(i)** 3.0476... **[M1]**

3.05 **[A1]**

**(ii)** 3 **[B1]**

**(iii)** Upper quartile = 4 and Lower quartile = 2 identified **[M1]**

2 **[A1]**

*[Total Marks: 5]*

**2** **(a)** $20 < t \leq 30$ **[B1]**

**(b)** Use of correct midpoints: $5, 15, 25, 35, 50$ **[M1]**

$\Sigma fx = 14(5) + 28(15) + 35(25) + 18(35) + 5(50)$ or 2245 seen **[M1]**

22.45 **[A1]**

**(c)** The exact times are not known so the midpoints of the intervals are used. **[B1]**

*[Total Marks: 5]*

**3** **(a)** Total frequency $= 10 + k + 2k + 3 + 7 + 5 = 3k + 25$ **[M1]**

Total score $= 10(1) + k(2) + (2k + 3)(3) + 7(4) + 5(5) = 8k + 72$ **[M1]**

$\dfrac{8k + 72}{3k + 25} = 2.8$ **[M1]**

$8k + 72 = 8.4k + 70 \implies 0.4k = 2 \implies k = 5$ **[A1]**

**(b)** **(i)** 3 **[B1]**

**(ii)** 1.5 **[B1]**

**(iii)** 4 **[B1]**

*[Total Marks: 7]*